

Yihao Zhou

McWilliams Center for Cosmology, Carnegie Mellon University
5000 Forbes Avenue, Pittsburgh, PA 15213, USA

✉ yihaoz@andrew.cmu.edu • [yihaozhou27.github.io](https://github.com/yihaozhou27)

RESEARCH INTERESTS

Cosmological hydrodynamic simulations • gravitational waves • dynamics of massive black holes • massive black hole seeding • AGN feedback • multi-messenger astronomy

EDUCATION

Carnegie Mellon University, Pittsburgh, PA Aug 2022 – present

Ph.D. Physics (expected 2027)

Advisor: Tiziana Di Matteo

Xi'an Jiaotong University, Xi'an, China Sep 2017 – May 2021

B.S. Physics (with honours)

Advisors: Jun Feng, Haiguang Xu

Thesis: *A Semi-Analytical Model for the Formation and Evolution of Radio Relics in Galaxy Clusters*

HONOURS AND AWARDS

NASA FINESST Fellowship (\$150,000) 2025–2028

NASA Goddard Travel Grant (\$1,500), 16th LISA Symposium June 2026

Outstanding Graduate, Xi'an Jiaotong University 2021

Exchange Program Scholarship for Excellent Undergraduate, Xi'an Jiaotong University 2019

Siyuan Scholarship, Xi'an Jiaotong University 2018–2021

PUBLICATIONS

First-Author Publications

Y. ZHOU, A. K. Bhowmick, T. Di Matteo, P. LaChance, R. Croft, L. Blecha et al., *First results of AMBRA: Abundant Seeds and Early Mergers as a Pathway to the First Massive Black Holes*, *arXiv e-prints* (2026) [2604.01123]

Y. ZHOU, T. Di Matteo, S. Bird, R. Croft, Y. Ni, Y. Yang et al., *The ASTRID Simulation at $z = 0$: From Massive Black Holes to Large-scale Structure*, *ApJ* **999** (2026) 41 [2510.13976]

Y. ZHOU, T. Di Matteo, N. Chen, L. Z. Kelley, L. Blecha, Y. Ni et al., *Central Cluster Galaxies: A Hot Spot for Detectable Gravitational Waves from Black Hole Mergers*, *ApJ* **988** (2025) L74 [2502.01845]

Y. ZHOU, D. Mukherjee, N. Chen, T. D. Matteo, P. H. Johansson, A. Rantala et al., *MAGICS. II. Seed Black Holes Stripped of Their Surrounding Stars Do Not Sink*, *ApJ* **980** (2025) 79

Y. ZHOU, H. Chen, T. D. Matteo, Y. Ni, R. A. C. Croft and S. Bird, *Modeling quasar proximity zones in a realistic cosmological environment with a self-consistent light curve*, *MNRAS* (2024) stae172

Y. ZHOU, H. Xu, Z. Zhu, Y. Zhao, S. Fan, C. Shan et al., *A semi-analytical model for the formation and evolution of radio relics in galaxy clusters*, *MNRAS* **517** (2022) 1299

Y. ZHOU and P. L. Plante, *Understanding the impact of semi-numeric reionization models when using cnns*, *PASP* **134** (2022) 044001

Publications with Major Contributions

N. Chen, **Y. ZHOU**, E. Dadiani, T. Di Matteo, C. Wang, A. Palmese et al., *Connecting current and future dual AGN searches to LISA and PTA gravitational wave detections*, *arXiv e-prints* (2025) [2512.16844]

B. Y. Wang, **Y. ZHOU**, W. Chen, N. Chen, T. Di Matteo, R. Croft et al., *Gravitational Waves from Massive Black Hole Mergers in ASTRID: Predictions for LISA*, *ApJ* **993** (2025) 199 [2503.24304]

- D. Mukherjee, **Y. ZHOU**, N. Chen, U. N. Di Carlo and T. Di Matteo, *MAGICS. III. Seeds Sink Swiftly: Nuclear Star Clusters Dramatically Accelerate Seed Black Hole Mergers*, *ApJ* **981** (2025) 203 [2409.19095]
- E. Dadiani, A. Palmese, **Y. ZHOU**, N. Chen, T. Di Matteo, A. ErósteGUI et al., *Cosmic Pairs: A DESI Census of Dual and Offset AGN as Precursors to Massive Black Hole Binaries*, *arXiv e-prints* (2026) [2607.27390]
- Y. Yang, S. Bird, **Y. ZHOU**, T. Di Matteo, R. Croft, Y. Ni et al., *Matter Clustering in Astrid: Reduced Baryonic Suppression from Realistic Black Hole Dynamics*, *arXiv e-prints* (2026) [2605.04176]
- N. Chen, T. Di Matteo, **Y. ZHOU**, L. Z. Kelley, L. Blecha, Y. Ni et al., *The Gravitational-wave Background from Massive Black Holes in the ASTRID Simulation*, *ApJ* **991** (2025) L19 [2502.01024]
- Y. Ni, N. Chen, **Y. ZHOU**, M. Park, Y. Yang, T. Di Matteo et al., *The Astrid Simulation: Evolution of Black Holes and Galaxies to $z = 0.5$ and Different Evolution Pathways for Galaxy Quenching*, *ApJ* **990** (2025) 120 [2409.10666]

Other Co-authored Papers

- F. Hafezianzadeh, T. Zhang, P. Rogozenski, P. Lachance, **Y. ZHOU**, T. Di Matteo et al., *The Galaxy Luminosity Functions in ASTRID: Predictions for LSST*, *arXiv e-prints* (2026) [2605.13843]
- D. Izquierdo-Villalba, M. Habouzit, M. Bonetti, S. Bonoli, A. Gualandris, M. Volonteri et al., *The LISA Astrophysics MBHcatalogues Project: A comparison of predictions of simulated massive black hole binaries*, *arXiv e-prints* (2026) [2605.00092]
- P. LaChance, A. K. Bhowmick, R. A. C. Croft, T. Di Matteo, **Y. ZHOU**, F. Pacucci et al., *From ASTRID to BRAHMA – The role of overmassive black holes in little red dots in cosmological simulations*, *arXiv e-prints* (2025) [2512.13957]
- P. LaChance, R. A. C. Croft, T. D. Matteo, **Y. ZHOU**, F. Pacucci, Y. Ni et al., *The Properties of Little Red Dot Galaxies in the ASTRID Simulation*, *The Open Journal of Astrophysics* **9** (2026) 55493 [2505.20439]
- R. Croft, P. Shaw, A.-M. Alexis, N. Chen, **Y. ZHOU**, T. D. Matteo et al., *Large-scale surveys of the quasar proximity effect*, *The Open Journal of Astrophysics* **8** (2025) 124 [2504.03848]
- J. Feng, J.-J. Zhang and **Y. ZHOU**, *Thermality of the unruh effect with intermediate statistics*, *Europhysics Letters* **137** (2022) 60001

PRESENTATIONS

- | | |
|---|-----------|
| The 16th LISA Symposium , University of Maryland | June 2026 |
| Poster: LISA Sources from Cosmological Simulations: MBH Mergers in ASTRID and AMBRA | |
| Bridging Scales Workshop , Black Hole Initiative, Harvard University | May 2026 |
| Talk: Building the First Massive Black Holes: Improving Seeding Models in Cosmological Simulations | |
| The 8th Neighborhood Workshop , Penn State University | Apr 2026 |
| Talk: From Nano- to Milli-Hz: Predicting Gravitational Wave Signals with the ASTRID/AMBRA Simulation | |
| Texas Symposium on Relativistic Astrophysics , Arizona State University | Dec 2025 |
| Talk: From Nano- to Milli-Hz: Predicting PTA and LISA Signals with the ASTRID Simulation | |
| Gravity in the Local Group , Carnegie Mellon University | Jun 2025 |
| Poster: The Seed Massive Black Hole Mergers in Dwarf Galaxies: What Can We Learn about the LISA Sources from the Cosmological Simulations ASTRID and MAGICS | |
| Ten Years to LISA , Jet Propulsion Laboratory | Apr 2025 |
| Talk: Understanding Massive Black Hole Seed Mergers: Insights from the MAGICS Simulations and Implications for LISA Observations | |
| Aspen 2025 Winter Conference , Aspen Center for Physics | Feb 2025 |
| Poster with flash talk: Central Cluster Galaxies: A Hotspot for Detectable Gravitational Waves from Black Hole Mergers | |
| Frontera User Meeting , Texas Advanced Computing Center | Aug 2024 |

Talk: Gravitational Waves Emitted by Massive Black Holes: Predictions from the Cosmological Simulation
ASTRID

Cosmology Seminar, Max Planck Institute for Astrophysics (remote) July 2024

Talk: MAGICS: The Crucial Role of Tidal Stripping in Seed Black Hole Binary Evolution

European Astronomical Society Annual Meeting, Padova, Italy (remote) July 2024

ePoster with short presentation: Modeling Quasar Proximity Zones in a Realistic Cosmological Environment with a Self-Consistent Light Curve

Galaxies Group Talk, University of Michigan (remote) Feb 2024

Talk: Modeling Quasar Proximity Zones with Self-Consistent Light Curves and Cosmological Environment

SAZERAC Specialist Session: Learning the High-Redshift Universe, remote Feb 2022

Talk: Understanding the Impact of Semi-Numeric Reionization Models when Using CNNs

4th Global 21-cm Workshop, University of Colorado Boulder (remote) Oct 2021

Poster: Understanding the Impact of Semi-Numeric Reionization Models when Using CNNs

PRESS

CMU Newsletter: *Ph.D. Student Wins NASA FINESST Grant* Dec 2025

SERVICE AND OUTREACH

Referee

Monthly Notices of the Royal Astronomical Society (MNRAS)

The Open Journal of Astrophysics (OJAp)

Membership

Laser Interferometer Space Antenna (LISA) Astrophysics Working Group

Conference Service

Local Organising Committee, *Gravity in the Local Group*, Carnegie Mellon University Jun 2025

Public Outreach

Astronomy on Tap, *Pittsburgh* — Supermassive Black Holes in the Baby Universe July 2027

MENTORING AND TEACHING

Student Advising

Fatemeh Hafezianzadeh (Graduate student, CMU)

Abdulla Alhosani (Graduate student, CMU)

Bonny Wang (Master student, CMU)

Cici Wang (Undergraduate student, CMU)

William Chen (Undergraduate student, CMU)

Ines Rodriguez Hsu (Undergraduate student, CMU)

Teaching Assistant

Physics I for Science Students, *Carnegie Mellon University* Fall 2023

Physics I for Engineering Students, *Carnegie Mellon University* Spring 2023

Physics I for Science Students, *Carnegie Mellon University* Fall 2022